



School of Computing Science and Engineering

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Project Title:	Implementation of AES, DES, and RSA for Secure Text Encryption	
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ABSTRACT

In the modern digital era, securing data during transmission and storage is essential. This project presents a secure text encryption and decryption system using three widely adopted cryptographic algorithms: Advanced Encryption Standard (AES), Data Encryption Standard (DES), and Rivest–Shamir–Adleman (RSA). The system provides a user-friendly interface that allows users to input plain text, select an algorithm, and perform encryption or decryption efficiently. AES and DES are symmetric key algorithms that ensure fast data processing, while RSA, an asymmetric algorithm, enhances security through public and private key pairs.

The application is developed with a frontend using HTML, CSS, and JavaScript for an interactive experience, while the backend handles cryptographic operations. By integrating multiple algorithms into a single platform, the system enables comparison of encryption techniques in terms of security and performance. This project highlights the importance of data confidentiality and serves as a foundational model for secure communication systems with applications in secure messaging, data protection, and cybersecurity.

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1. Introduction

With the rapid growth of digital communication, ensuring the security of data has become a major concern. Sensitive information such as personal messages, financial data, and confidential documents are frequently transmitted over networks, making them vulnerable to unauthorized access and cyber attacks. Cryptography plays a vital role in protecting such data by converting readable information (plaintext) into an unreadable format (ciphertext).

This project focuses on developing a secure text encryption and decryption system using three standard cryptographic algorithms: Advanced Encryption Standard (AES), Data Encryption Standard (DES), and Rivest–Shamir–Adleman (RSA). These algorithms are widely used in modern security systems to ensure confidentiality, integrity, and authenticity of data.

The system provides a simple and interactive interface where users can input text, choose an encryption algorithm, and perform encryption or decryption operations. By integrating both symmetric (AES, DES) and asymmetric (RSA) encryption techniques, the project demonstrates different approaches to securing data and highlights their practical applications.

This project serves as a foundational step towards understanding cybersecurity principles and implementing secure communication systems in real-world scenarios.

2. Objective

- To design and develop a secure text encryption and decryption system.
- To implement and demonstrate the working of AES, DES, and RSA algorithms.
- To understand the difference between symmetric and asymmetric encryption techniques.
- To provide a user-friendly interface for performing encryption and decryption operations.
- To ensure data confidentiality during transmission and storage.
- To gain practical knowledge of cryptographic techniques used in cybersecurity.

3. Theory

Cryptography is the process of securing data by converting plaintext into ciphertext to prevent unauthorized access. It ensures confidentiality and data protection during communication.

1. Encryption and Decryption

Encryption converts readable data into an unreadable form using an algorithm and a key. Decryption converts it back to its original form.

2. Types of Cryptography

- Symmetric Key Cryptography: Uses the same key for encryption and decryption. It is fast and efficient.
Examples: AES, DES
- Asymmetric Key Cryptography: Uses two keys (public and private). It is more secure but slower.
Example: RSA

2. AES (Advanced Encryption Standard)

AES is a secure symmetric algorithm that uses key sizes of 128, 192, or 256 bits and is widely used in modern systems.

4. DES (Data Encryption Standard)

DES is an older symmetric algorithm with a 56-bit key. It is less secure compared to AES.

5. RSA (Rivest–Shamir–Adleman)

RSA is an asymmetric algorithm based on mathematical principles. It uses a public key for encryption and a private key for decryption.

Cryptography is widely used in secure communication, banking systems, and data protection applications

4. Methodology

The methodology of this project focuses on designing a system that securely encrypts and decrypts text using multiple cryptographic algorithms. The overall process is carried out in the following steps:

1. **User Input:**
The user enters the plaintext or ciphertext into the input field provided in the interface.
2. **Algorithm Selection:**
The user selects one of the available encryption algorithms (AES, DES, or RSA) from the dropdown menu.
3. **Request Processing:**
Based on the selected operation (encryption/decryption), the input data is sent to the backend server through HTTP requests.
4. **Encryption/Decryption Process:**
The backend applies the selected algorithm to transform the data into ciphertext (encryption) or plaintext (decryption).
5. **Result Generation:**
The processed output is generated and returned to the frontend.

Display Output:

The result is displayed to the user in the output section of the interface

5. Implementation

The project is implemented using:

Frontend:

Built using HTML, Tailwind CSS, and JavaScript to provide an interface for user input, algorithm selection, and encryption/decryption operations.

Backend(JavaScript):

Implemented using Node.js and Express.js to handle API requests (/encrypt, /decrypt) and perform cryptographic operations.

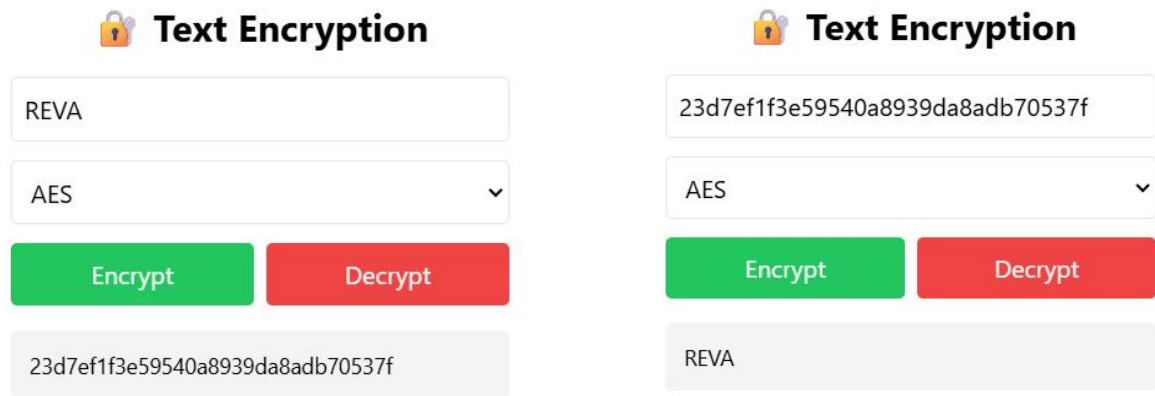
Algorithms:

AES and DES are implemented as symmetric encryption techniques, while RSA uses public and private keys for secure encryption and decryption.

Communication:

The frontend and backend communicate using JavaScript through REST APIs in JSON format.

6. OUTPUT



Text Encryption

REVA

AES

Encrypt Decrypt

23d7ef1f3e59540a8939da8adb70537f

Text Encryption

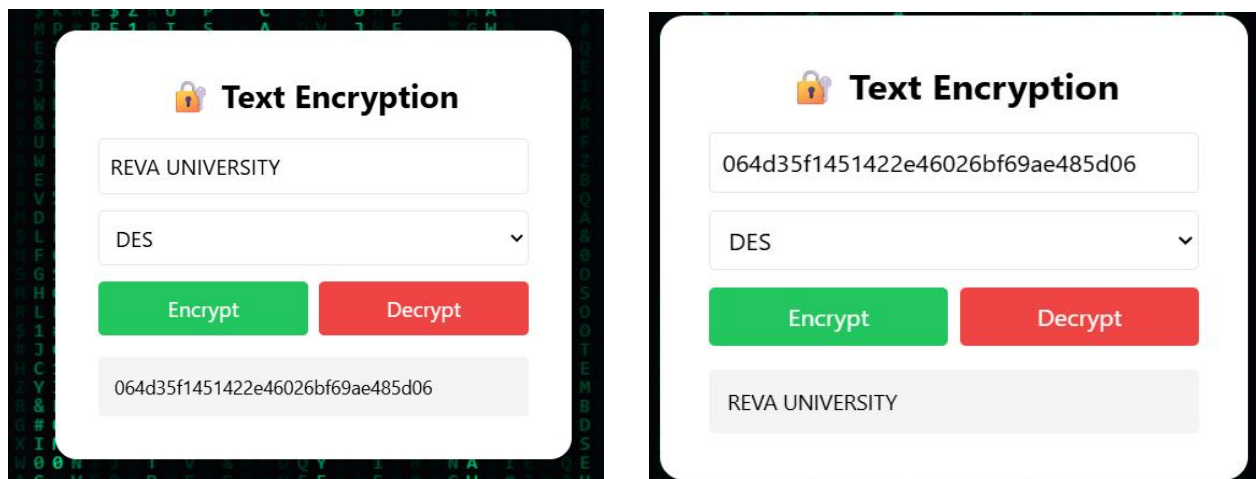
23d7ef1f3e59540a8939da8adb70537f

AES

Encrypt Decrypt

REVA

Figure 1: Text Encryption using AES Algorithm



Text Encryption

REVA UNIVERSITY

DES

Encrypt Decrypt

064d35f1451422e46026bf69ae485d06

Text Encryption

064d35f1451422e46026bf69ae485d06

DES

Encrypt Decrypt

REVA UNIVERSITY

Figure 2 : Text Encryption using DES Algorithm

Text Encryption

REVA

RSA

Encrypt

Decrypt

39ef7fbc6685ffdf102ccb54f16c16e9bebd78603f
dc9a90939272795b35ba2477d4caafc50f2190fc
e67df2e8e4f24c1afe55116b79440d7df5bcd9f38
81592c027468a29e1df7538cbdb6597ff7f6035b
099425499e2c8e4fa11103f36feaca761597f001d
673b503b29bdbf5a80da0b546900e3fbd8ff839b
5393d233a58d0f5245755f375822a684b29473a
be683ec7260f8d1e4205ea08907cfab97ed0ea99
701ec0c26af6ae0eb3e230ff7d1ade226cab9784
e4b9f51fa5d3752db3ee597736844b8a3827017
9c5a7779d9ad3fa27e77763b4e2e7b1f88105309
4bf7edb9844dbef3bae36152f1cb671fedf4a868
864eeae450189ef1107657062f7bb6



Text Encryption

39ef7fbc6685ffdf102ccb54f16c16e9bebd78

RSA

Encrypt

Decrypt

REVA

Figure 3: Text Encryption using RSA Algorithm

7. Advantages

- Provides data security by converting plaintext into ciphertext
- Supports multiple encryption algorithms (AES, DES, RSA)
- Helps in understanding both symmetric and asymmetric encryption
- Combines Symmetric (AES, DSA) and Asymmetric (RSA) algorithms for both speed and security

8. Limitations

- DES is less secure due to its small key size.
- RSA is slower compared to symmetric algorithms.
- Security depends on proper key management.
- Limited to text-based encryption only.

9.Applications

- Data security in web applications
- Password protection and authentication
- Secure messaging systems
- Cryptographic protocols

10. Future Enhancements

- Improve security by implementing stronger algorithms.
- Add support for file and image encryption
- Add user authentication and access control
- Improve key management and key generation features

11.Conclusion

This project successfully demonstrates the implementation of a secure text encryption and decryption system using AES, DES, and RSA algorithms. It highlights the importance of cryptography in protecting data from unauthorized access. By integrating both symmetric and asymmetric encryption techniques, the system provides a clear understanding of different security approaches. The use of JavaScript for both frontend and backend makes the system efficient and easy to use. Overall, the project serves as a foundational model for secure communication and data protection systems.

12. Project Links

The project source code can be accessed through the following links:

GitHub Repository:

<https://github.com/SakthiRishikesh/text-encryption-system>

13. References

- William Stallings, *Cryptography and Network Security*
- GeeksforGeeks, “RSA Algorithm.” <https://www.geeksforgeeks.org/rsa-algorithm-cryptography/>
- GeeksforGeeks, “DES Algorithm.” <https://www.geeksforgeeks.org/data-encryption-standard-des-/>
- GeeksforGeeks, “AES Algorithm.” <https://www.geeksforgeeks.org/advanced-encryption-standard-/>
- NIST (National Institute of Standards and Technology), Cryptographic Standards and Guidelines.
<https://www.nist.gov>
- Lecture Notes and Course Materials – Computer Networks and Security (CNS).